1 Model

We develop a static general equilibrium model of multinational production in which the strategic agent is the ownership group. A group can operate plants in several countries and, when it sets quantities, takes into account that its own plants compete with one another. Markups therefore depend on the group's total share of the market rather than on the share of any single plant. Profits are paid to the countries that own each group, and ownership is recorded group by group rather than country by country. The baseline takes as given which plants each group operates and which markets they serve; Section 1.5 lets groups choose them. Proofs are in the Appendix.

1.1 Model Set-Up

Agents. There is a finite set of countries $\mathcal{N} = \{1, \dots, N\}$ and a finite set of sectors $\mathcal{K} = \{1, \dots, K\}$. Each country n is inhabited by a representative household that supplies L_n units of labor inelastically and owns shares in firms. Production takes place in plants, and every plant belongs to exactly one group $g \in \mathcal{G}$, which we identify with its global ultimate parent. Each group has a headquarters country $h(g)$, operates in a single sector $k(g)$,[1] and owns a finite set of plants $\mathcal{P}_g$. We index plants by i and write $g(i)$ and $\ell(i)$ for the group and the location of plant i. Each plant produces one differentiated variety. A *market* is a destination–sector pair (n, k). Plant i serves a given set of destinations, and $\mathcal{A}_{nk}$ denotes the set of plants that serve market (n, k). A group with plants in more than one country is a multinational, a plant located outside its group's headquarters country is a foreign affiliate, and a group with a single plant is a stand-alone firm. We assume that every market is served by at least two groups.

Representative household. The household in country n has Cobb–Douglas preferences across sectors and CES preferences within them,

$$U_n = \prod_{k \in \mathcal{K}} C_{nk}^{\beta_k}, \qquad C_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} q_{i,n}^{(\sigma_k - 1)/\sigma_k} \Big]^{\sigma_k/(\sigma_k - 1)}, \tag{1}$$

where $\beta_k > 0$, $\sum_k \beta_k = 1$, and $\sigma_k > 1$. The household spends the share β_k of its income X_n on sector k, so spending on market (n, k) is

$$E_{nk} = \beta_k X_n, \tag{2}$$

and demand for the variety of plant i is

$$q_{i,n} = p_{i,n}^{-\sigma_k} \, P_{nk}^{\sigma_k - 1} E_{nk}, \qquad P_{nk} = \Big[\sum_{i \in \mathcal{A}_{nk}} p_{i,n}^{1 - \sigma_k} \Big]^{1/(1 - \sigma_k)}. \tag{3}$$

Welfare is real income X_n / P_n, with $P_n = \prod_k (P_{nk}/\beta_k)^{\beta_k}$.[2]

[1] This restriction is without loss for behavior. Marginal costs are constant, so a group's decisions separate across markets (Section 1.2). A group active in several sectors is equivalent to several single-sector groups with the same owners; what the restriction rules out is a capability shared across sectors.

[2] With Cobb–Douglas preferences across sectors, a sector's spending does not expand when its price index falls. A CES upper tier with elasticity $\eta \in (1, \sigma_k)$ would add that margin and change the markup in (8) to $1/\mu(S) = 1 - (1 - S)/\sigma_k - S/\eta$; the structure of the market equilibrium below is unchanged.

Production technology. Labor is the only factor of production. A plant located in ℓ produces with constant returns, and its unit cost is

$$a_i = \frac{w_\ell}{\varphi_i}, \tag{4}$$

where w_ℓ is the wage and $\varphi_i > 0$ the plant's productivity.[3] Productivity can combine a component of the location, a component of the plant, and a component common to all plants of the group, which captures head-office services such as management, technology and brands (Antràs, 2003; Antràs and Helpman, 2004). These services are a non-rival capability of the group that all of its plants share, so they enter productivity rather than the cost of inputs.[4] A plant may also be less productive when it operates away from its group's headquarters (Ramondo and Rodríguez-Clare, 2013; Head and Mayer, 2019). The parametric form of φ_i matters for the quantification, not for the results of this section.

Trade. Shipping goods from ℓ to n entails an iceberg cost $d_{\ell n} \geq 1$, with $d_{nn} = 1$. Trade costs run from the country where the good is produced, not from headquarters, so a single plant can serve several destinations, including third markets, as an export platform (Tintelnot, 2017). Country n levies a tariff $t_{\ell nk} \geq 0$ on sector-k goods produced in ℓ, with $t_{nnk} = 0$. Tariffs depend on where a good is produced, not on who owns the plant that produces it, so a country's tariffs also fall on the foreign affiliates of the groups its residents own. We model the tariff as a proportional wedge on the cost of delivering the good. The delivered marginal cost of plant i in market (n, k) is

$$c_{i,n} = a_i \, d_{\ell n} \, (1 + t_{\ell nk}), \qquad \ell = \ell(i), \tag{5}$$

and country n collects $t_{\ell nk} \, a_i d_{\ell n} q_{i,n}$ on the sales of plant i.[5]

Profit repatriation. Country n owns a share $\theta_{gn} \geq 0$ of group g, with $\sum_n \theta_{gn} = 1$. Groups maximize their own profits whoever owns them: ownership determines where profits are spent, not how groups price. The profit of a group, $\pi_g = \sum_n \pi_{g,n}$, where $\pi_{g,n}$ is its profit in market $(n, k(g))$ (Section 1.2), is paid to its owners in proportion to θ_{gn}, wherever it is earned. Tariff revenue is rebated lump sum to the household of the country that collects it. Income in country n is

$$X_n = w_n L_n + \sum_{g \in \mathcal{G}} \theta_{gn} \, \pi_g + T_n, \qquad T_n = \sum_k \sum_{i \in \mathcal{A}_{nk}} t_{\ell(i)nk} \, a_i \, d_{\ell(i)n} \, q_{i,n}. \tag{6}$$

Profits are earned where groups sell and received where their owners reside, and θ links the two. A natural benchmark assigns each group to the country of its ultimate parent, $\theta_{gn} = \mathbf{1}\{n = h(g)\}$; nothing below requires this form. Because income includes the profits of the groups a country owns, wherever they produce and sell, trade need not balance country by country.

1.2 Market Structure

Fix a market (n, k) and suppress its indices: we write σ, E and P for σ_k, E_{nk} and P_{nk}, c_i for the delivered cost $c_{i,n}$, and T_g for the plants of group g that serve the market.

[3] We do not include intermediate inputs. With an input–output structure, a tariff also raises the cost of the input bundle, and the uniqueness and existence results below require additional conditions.

[4] The common alternative treats head-office services as labor bought at the parent's wage, so that a plant abroad pays part of its costs at a foreign wage. This links factor markets across countries and can overturn the substitutability of labor across countries on which uniqueness of wages rests.

[5] A duty on the exporter's price yields the same prices when all the plants a group uses to serve n face the same tariff. When they do not, it makes the group's markup differ across its plants, and the property that a group's markup depends only on its total share (Section 1.2) is lost.

Demand. Let $A \equiv \sum_{i \in \mathcal{A}} p_i^{1-\sigma} = P^{1-\sigma}$. By (3), the revenue share of plant i and the share of group g are

$$s_i = \frac{p_i q_i}{E} = \frac{p_i^{1-\sigma}}{A}, \qquad S_g = \sum_{i \in T_g} s_i, \qquad \sum_g S_g = 1. \tag{7}$$

The aggregate A summarizes how competitive the market is: it is large when rivals charge low prices, and then any given price wins a small share.

Competition. Groups are the strategic agents. Each group chooses the quantities of all of its plants that serve the market, taking as given the quantities of other groups and the level of spending E. It maximizes its profit $\pi_g = \sum_{i \in T_g} (p_i - c_i) q_i$.

Proposition 1 (Markups)**.** *The first-order conditions of group g are $p_i = \mu(S_g)\, c_i$ for every $i \in T_g$, with*

$$\mu(S) = \frac{\sigma}{(\sigma - 1)(1 - S)}. \tag{8}$$

This is the markup of Atkeson and Burstein (2008), evaluated at the share of the group rather than of the plant.[6] It rises from $\sigma/(\sigma-1)$ for a negligible group and is unbounded as the share tends to one, which is why every market needs at least two groups.

Cannibalization across multinational affiliates. When a group expands one of its plants, the sector price index falls and every variety in the market loses sales, the group's other plants included. A single-plant firm ignores the losses it imposes on others; a group counts the losses it imposes on itself, and these grow with its share. This is the factor $1 - S_g$ in (8). Two implications follow. First, all plants of a group charge the same markup in a given market, and differences in their costs show up in quantities rather than in markups. Second, market power is a property of the group: concentration should be measured at the level of groups, and a group charges more than its plants would charge as independent firms. Within a group, (7) and the common markup give

$$\frac{s_i}{S_g} = \frac{c_i^{1-\sigma}}{K_g}, \qquad K_g \equiv \sum_{i \in T_g} c_i^{1-\sigma}. \tag{9}$$

We call K_g the *strength* of the group in the market: an index of its cost efficiency that rises with the number of plants serving the market and with how cheaply each delivers. Substituting $p_i = \mu(S_g) c_i$ into (7),

$$\psi(S_g) = \frac{K_g}{A}, \qquad \psi(S) \equiv S\, \mu(S)^{\sigma-1} = \Big(\frac{\sigma}{\sigma-1}\Big)^{\sigma-1} \frac{S}{(1-S)^{\sigma-1}}. \tag{10}$$

The map ψ is strictly increasing, so $S_g = \psi^{-1}(K_g/A)$: the share of a group depends only on its strength relative to the competitiveness of the market. However many plants a group runs, and in however many countries, its position in the market is summarized by one number, K_g, and all it needs to know about its rivals is A. The equilibrium of the market is the value of A at which shares add up to one,

$$\sum_g \psi^{-1}\left(\frac{K_g}{A}\right) = 1. \tag{11}$$

[6] Under price competition the markup is $\varepsilon(S)/[\varepsilon(S)-1]$ with $\varepsilon(S) = \sigma - (\sigma-1)S$. It is also common to the group's plants and increasing in its share, so the structure below carries over, but profits are no longer quadratic in the share and the ownership result of Section 1.4 becomes an approximation.

Profits. Since $c_i q_i = p_i q_i / \mu(S_g)$, the profit of group g is $\pi_g = E\, S_g\, [1 - 1/\mu(S_g)]$, and (8) makes the Lerner index linear in the share:

$$\pi_g = \frac{E}{\sigma}(S_g + \kappa S_g^2), \qquad \kappa \equiv \sigma - 1. \tag{12}$$

Profits rise more than proportionally with the share, because a larger group both sells more and earns more on each unit sold. The curvature κ governs the ownership results of Section 1.4. It would be zero with constant markups.

1.3 General Equilibrium

Labor market. Let $V_{i,n'} \equiv a_i d_{\ell(i)n'} q_{i,n'}$ be the cost of plant i in serving country n', exclusive of tariffs. By Proposition 1, $V_{i,n'} = r_{i,n'}/[\mu_{g(i)}(1 + t_{\ell(i)n'k})]$, where $r_{i,n'}$ is the plant's revenue in that market and $\mu_{g(i)}$ its group's markup there. All of this cost is paid to labor where the plant is located, so labor markets clear when

$$w_n L_n = \sum_{k \in \mathcal{K}} \sum_{n' \in \mathcal{N}} \sum_{i \in \mathcal{A}_{n'k}:\, \ell(i) = n} V_{i,n'}. \tag{13}$$

Definition 1 (Equilibrium)**.** An equilibrium consists of wages $\{w_n\}$, incomes $\{X_n\}$, and prices and quantities $\{p_{i,n}, q_{i,n}\}$ such that households maximize (1); in every market, prices satisfy Proposition 1 and quantities satisfy (3); incomes satisfy (6), with market sizes (2); and labor markets clear, (13).

Prices, costs and incomes are homogeneous of degree one in wages, and shares are homogeneous of degree zero, so wages are determined up to scale. Adding (6) across countries shows that world income equals world spending, and one labor-market condition is implied by the others (Walras' law). We set $w_1 = 1$.

Existence and uniqueness. We establish the results layer by layer: the equilibrium of a single market, incomes given wages, and wages.

Proposition 2 (One market)**.** *For any positive delivered costs, a market has exactly one equilibrium.*

The proof uses (11): each $\psi^{-1}(K_g/A)$ falls as A rises, so the sum of shares is a strictly decreasing function of a single number and equals one exactly once. Each group's profit is strictly concave in its own quantities, so the first-order conditions identify the unique best response.

Proposition 3 (Incomes)**.** *Given wages, incomes are the unique solution of the linear system* (2)*,* (6)*.*

Given wages, every share and markup is known, profits and tariff revenue are linear in market sizes, and market sizes are linear in incomes. The system has a unique solution because each dollar a country spends returns to the world as profits and tariffs by less than a dollar: the rest is paid to labor.

Proposition 4 (Existence of wages)**.** *Suppose no set of countries is cut off from the rest: for every proper subset Z of countries, some plant located in Z serves a market of a country outside Z. Then there exist wages that clear every labor market.*

Uniqueness is the most delicate layer. Let $W_n(w)$ be the right side of (13), the wage bill of country n at wages w, and $M_{nj} \equiv \partial \ln W_n / \partial \ln w_j$. Homogeneity makes every row of M sum to one. Labor in different countries is a gross substitute when $M_{nj} > 0$ for $n \neq j$: a rise in any other country's wage raises country n's wage bill. Let $b^n_{i,n'}$ be the share of country n's wage bill paid by plant i out of its costs in country n', and

$$\bar{E}^{(j)}_n \equiv \sum_{i,n'} b^n_{i,n'} \frac{\partial \ln E_{n'k(i)}}{\partial \ln w_j}, \tag{14}$$

the average response of the markets in which country n's plants sell.

Theorem 1 (Wages)**.** *Let D be a set of wage vectors that is convex in logarithms. Suppose that at every $w \in D$ (i) no group holds more than half of any market and (ii) $\bar{E}^{(j)}_n > 0$ for all $n \neq j$. Then $M_{nj} \geq \bar{E}^{(j)}_n > 0$ for $n \neq j$ throughout D, and D contains at most one equilibrium wage vector up to scale.*

The bound $M_{nj} \geq \bar{E}^{(j)}_n$ says that a rise in a foreign wage can lower country n's wage bill only by shrinking the markets in which its plants sell; every reallocation of sales between plants and between groups favors country n. Condition (i) is what signs the effect of variable markups on labor demand. With gross substitutes, a second equilibrium is ruled out by the classical argument of Arrow, Block, and Hurwicz (1959). Condition (ii) is not a theorem of the model. A country whose groups lose profits when a foreign wage rises spends less, and if the markets where country n sells shrink enough, its wage bill can fall. That income effect is the ownership channel the model is built to study. Both conditions are inequalities in objects the model computes, and we report them with the quantification.

Solution algorithm. Given wages, each market is solved for A from (11) by safeguarded Newton steps on $\ln A$, which never leave a bracket that contains the unique root, and incomes come from one linear solve. Wages are updated by the damped tâtonnement $w_n \leftarrow w_n^{1-\lambda}(W_n(w)/L_n)^{\lambda}$, which raises a country's wage when its labor is in excess demand and whose fixed points are the equilibrium wages, followed by Newton steps once close to the solution. Labor demand is elastic, so M_{nn} can be negative and an undamped update can overshoot; we choose λ with $\lambda(1 - M_{nn}) < 1$, which keeps the update monotone.

1.4 Comparative Statics

Incidence within a market. Consider a market with given plants and spending, and a change in delivered costs. Let $\hat{c}_g = \sum_{i \in T_g} (s_i / S_g)\, d\ln c_i$ be the average cost change of group g, and

$$\Lambda_g \equiv \frac{1 - S_g}{1 + (\sigma - 2) S_g}, \qquad \varpi_g \equiv \frac{S_g \Lambda_g}{\sum_h S_h \Lambda_h}. \tag{15}$$

Λ_g falls from one toward zero as the share rises.

Proposition 5 (Incidence)**.** *Holding spending fixed,*

$$d\ln S_g = -(\sigma - 1)\Lambda_g (\hat{c}_g - d\ln P), \qquad d\ln P = \sum_g \varpi_g\, \hat{c}_g. \tag{16}$$

If only group j's costs change, its sales-weighted average price rises by the fraction $\rho_j = 1 - (\sigma - 1) S_j \Lambda_j (1 - \varpi_j)/(1 - S_j) \in (0, 1]$ of its average cost increase $\hat{c}_j$, and every rival h gains share, $d\ln S_h = (\sigma - 1)\Lambda_h \varpi_j \hat{c}_j$, and raises its price.

In the limit of small groups, which is monopolistic competition, $\Lambda_g = 1$, $\varpi_g = S_g$ and $\rho_j = 1$. Large groups instead absorb part of a cost increase in their markup, so pass-through falls with the group's share, and since $\varpi_g < S_g$ for the larger groups, the price index moves by less than their share would suggest. Rivals are not passive either: a cost increase for one group raises their shares, markups and prices. Because tariffs depend on where goods are produced, a tariff on goods from ℓ raises $\hat{c}_g$ only by the share of group g's sales that comes from its plants in ℓ: a group able to serve the market from several countries is partly shielded, while a stand-alone exporter from ℓ bears the tariff in full.

Ownership and the level of profit income. Consider a country H with ownership shares $\theta_g \equiv \theta_{gH}$, and its profit income from a market, $\pi^H = \sum_g \theta_g \pi_g$.

Proposition 6 (Ownership-weighted concentration)**.**

$$\frac{\pi^H}{E} = \frac{1}{\sigma}\Big[\bar{\theta} + \kappa\big(\bar{\theta} \cdot \mathrm{HHI} + \mathrm{Cov}_S(\theta, S)\big)\Big], \tag{17}$$

where $\bar{\theta} = \sum_g S_g \theta_g$ *is the sales-weighted ownership share,* $\mathrm{HHI} = \sum_g S_g^2$*, and* $\mathrm{Cov}_S(\theta, S) = \sum_g \theta_g S_g^2 - \bar{\theta} \cdot \mathrm{HHI}$ *is the covariance between ownership and share computed with sales weights. A calculation that applies* $\bar{\theta}$ *to total profit in the market,* $\bar{\theta} \sum_g \pi_g$*, misses exactly* $(E\kappa/\sigma)\,\mathrm{Cov}_S(\theta, S)$*.*

The first term is what a model with constant markups delivers: ownership times revenue over σ. The second is an ownership-weighted Herfindahl index. Its component $\bar{\theta} \cdot \mathrm{HHI}$ can be built from a country's ownership share and a concentration index; the covariance cannot, since it requires knowing which groups a country owns and how large each one is. Because profits are convex in shares, two countries with the same share of a market's sales earn different profits if one owns the large groups and the other the small ones. With constant markups, as under monopolistic competition, $\kappa = 0$ and a country's profit income is its ownership share times total profit, whatever the distribution of ownership across groups. Summed over markets with spending weights, the same algebra splits the economy-wide correction into a covariance across markets between ownership shares and concentration, and the average of the covariances within markets.

Ownership and the response of profit income. Welfare depends on how profit income responds to policy, and the response is governed by a different statistic.

Proposition 7 (Response of owned profits)**.** *Holding spending fixed, a change in the delivered costs of group* j *alone changes country* H*'s profit income by*

$$\frac{d\pi^H}{E} = \frac{\sigma - 1}{\sigma}\, \varpi_j\, \hat{c}_j \sum_{h \neq j} S_h \Lambda_h \Big[\theta_h(1 + 2\kappa S_h) - \theta_j(1 + 2\kappa S_j)\Big]. \tag{18}$$

A cost increase for group j alone, for example a tariff on goods from a country from which only j serves the market, reallocates sales from j to its rivals in proportion to $S_h \Lambda_h$, and each unit of share is worth $\theta(1 + 2\kappa S)$ to country H, up to the factor E/σ. The country gains when the rivals it owns are worth more to it at the margin than the group whose costs rose. The weights are $S_h \Lambda_h$ rather than S_h^2, so large groups count for less in the response than in the level, and what matters is the ownership of the affected group relative to its rivals, not the covariance that governs the level. The ownership-weighted Herfindahl index measures a country's stake in a market but is not a sufficient statistic for how that stake responds to a tariff; in general equilibrium the response also runs through wages and market sizes.

Wages and prices. A rise in country j's wage raises the delivered cost of every plant located in j one for one. By Proposition 5, it raises the price index of every market by $\sum_g \varpi_g \omega_{gj}$, where ω_{gj} is the share of group g's sales in the market that comes from its plants in j: a proportion between zero and one, so markups dampen a wage change but cannot reverse or amplify it. Who gains from a fall in j's wage depends on ownership. The fall lowers the costs of every group with plants in j, and part of the gain accrues as profit to the owners of those groups, wherever they reside. A policy that lowers a foreign wage can therefore raise the income of the country that owns the plants located there. This is the general-equilibrium channel through which ownership matters for trade policy.

1.5 Extension: Endogenous Entry

The baseline takes as given which plants each group operates and which markets they serve. This subsection lets groups choose them, so that which affiliates exist, and where each of them sells, is an equilibrium outcome.

Fixed costs and plant sets. Each group now has a set of potential plants $\mathcal{P}_g$. A plant that serves market (n, k) pays a fixed cost

$$F_{i,n} = f_k \, w_\ell \, f^d_{\ell n}, \tag{19}$$

in units of labor of the country where it is located, with $f_k > 0$ and $f^d_{\ell n} \geq 1$. The cost is incurred once per market served, whatever the quantity sold, and does not depend on who owns the plant. Because it is paid in labor, it appears as wage income and leaves the model homogeneous of degree one in wages. Every fixed cost is specific to a plant–market pair, so a group's problem separates across destinations: in each market $(n, k(g))$ it chooses the subset $T_{g,n} \subseteq \mathcal{P}_g$ of its plants that serve n.[7] This is a simplified version of the multi-plant location problem of Yang (2023).

The entry problem. A group's plant set matters for its profit only through its strength. Let $K(T) = \sum_{i \in T} c_i^{1-\sigma}$ and $\Phi(T) = \sum_{i \in T} F_{i,n}$. Given A, serving the market with T yields

$$U_g(T; A) = \Omega\big(K(T); A\big) - \Phi(T), \qquad \Omega(K; A) \equiv \frac{E}{\sigma}(S + \kappa S^2)\Big|_{S = \psi^{-1}(K/A)}. \tag{20}$$

Because Ω is strictly increasing in K, a set can be optimal only if no other set delivers at least as much strength at no greater fixed cost. These efficient sets form a chain, ordered by strength and by cost together, so the group's choice amounts to how far to climb a ladder that runs from small and cheap to large and expensive. When a group's plants pay the same fixed cost to serve the market, each rung adds the plant with the next-lowest delivered cost, and the group's choice is a cutoff on its own list of plants, the within-group analogue of the cutoff in Gaubert and Itskhoki (2021).

Definition 2 (Entry equilibrium in a market)**.** Given delivered costs, fixed costs and spending, an entry equilibrium is a collection of plant sets $\{T_g\}$ and an aggregate A such that (i) each T_g maximizes $U_g(\cdot\,; A)$ over subsets of $\mathcal{P}_g$, and (ii) A solves (11) with $K_g = K(T_g)$.

Groups evaluate entry at the prevailing degree of competition, as in a free-entry condition, while taking into account their effect on A when they set quantities. This notion turns each group's entry problem into the choice of one number given one number.

[7] A plant-level setup cost would link a group's choices across markets and turn its problem into a combinatorial one over locations and markets jointly (Tintelnot, 2017).

Theorem 2 (Entry)**.** *Let $\sigma \geq 2$.*

(a) $\Omega(K; A)$ is strictly concave in K at every share.

(b) $\Omega_{KA} < 0$ whenever the group's share is at most one half.

(c) Suppose that, for any two candidate equilibria, no group would hold more than half of the market if it operated the plants it chooses in one of them while facing the aggregate of the other, and that no group is indifferent between two rungs of its ladder. Then a market has at most one entry equilibrium.

Part (a) says that strength has diminishing returns for the group, and part (b) that it is worth less in a more competitive market. Together they imply that each group's optimal strength falls as A rises, while A rises with every group's strength, so two equilibria with different aggregates cannot both exist. No order of entry is imposed, and groups still internalize competition among their own plants. The hypothesis is the same one-half condition as in Theorem 1.

General equilibrium with entry. Fixed costs are paid to labor where the plant is located, so they are added to the right side of (13), and profits distributed through θ are net of them. The results of Section 1.3 hold for given plant sets, with one change: existence also requires fixed costs small enough that every country's income is at least a fixed fraction of its labor income. Two limits are new. First, plant sets change discretely with wages, so uniqueness at each layer does not make the joint equilibrium of wages and plant sets unique. Second, entry is chosen in whole plants, so an exact equilibrium in pure strategies need not exist. The solution algorithm adds an outer loop: at the equilibrium wages for the current plant sets, the groups with the largest gains from changing their sets do so, and the procedure repeats until no group wishes to move. Moving only the groups with the largest gains changes the path of the algorithm, not its fixed points, and does not amount to an order of entry. When the loop does not reach a configuration in which no group wishes to move, it returns the configuration with the fewest such groups.

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Appendix

A The Market Game

A.1 Proof of Proposition 1. Fix a market. Let $\rho = (\sigma-1)/\sigma$ and $Q = (\sum_i q_i^\rho)^{1/\rho}$. Since $PQ = E$, inverse demand is $p_i = E\,Q^{-\rho} q_i^{\rho-1}$, so $s_i = q_i^\rho/Q^\rho$ and group g's revenue is $R_g = E\,Q^{-\rho}\sum_{j\in T_g} q_j^\rho = E\,S_g$. Using $\partial \ln Q/\partial q_i = q_i^{\rho-1}/Q^\rho = s_i/q_i$,

$$\frac{\partial R_g}{\partial q_i} = E\,Q^{-\rho}\Big[\rho\, q_i^{\rho-1} - \rho\,\frac{q_i^{\rho-1}}{Q^\rho}\sum_{j\in T_g} q_j^\rho\Big] = \rho\, p_i\,(1-S_g).$$

Setting marginal revenue equal to c_i gives (8). The second term in the bracket is the loss the group imposes on its own plants. □

A.2 Proof of Proposition 2. *Best responses.* Given rivals' quantities, group g's profit is $E\,X_g/(X_g+B_g) - \sum_{i\in T_g} c_i q_i$, where $X_g = \sum_{i\in T_g} q_i^\rho$ and $B_g > 0$ is the same index for rivals. The function $X \mapsto X/(X+B)$ is strictly increasing and strictly concave, with derivatives $B/(X+B)^2 > 0$ and $-2B/(X+B)^3 < 0$, and X_g is strictly concave in $(q_i)_{i\in T_g}$. A strictly increasing, strictly concave function of a strictly concave function is strictly concave, and cost is linear, so profit is strictly concave in the group's own quantities. Since $\partial X_g/\partial q_i \to \infty$ as $q_i \to 0$, the optimum is interior and is the unique solution of the first-order conditions. *The share map.* By (10), $\psi(0) = 0$, ψ is continuous and strictly increasing on $[0,1)$, and $\psi(S) \to \infty$ as $S \to 1$, so it has a continuous, strictly increasing inverse defined on $[0,\infty)$. *The aggregate.* By Proposition 1 and (10), an equilibrium is an A solving (11), and conversely any such A generates prices $p_i = \mu(S_g)c_i$ that satisfy every group's first-order conditions and are consistent with A, since $\sum_i p_i^{1-\sigma} = \sum_g \mu(S_g)^{1-\sigma} K_g = A\sum_g S_g = A$. The left side of (11), $\mathcal{S}(A)$, is continuous and strictly decreasing. As $A \to \infty$ every share tends to zero, and as $A \to 0$ every share tends to one, so $\mathcal{S}$ tends to the number of groups, which is at least two. Hence $\mathcal{S}(A) = 1$ has exactly one solution. □

B General Equilibrium

B.1 Proof of Proposition 3. Given wages, delivered costs are known, and by Proposition 2 so are all shares and markups. Profits in market (n',k) are $E_{n'k}S_{g,n'}m(S_{g,n'})$, with $m(S) = (1+\kappa S)/\sigma$, and tariff revenue is $\tau_{n'k}E_{n'k}$, with $\tau_{n'k} = \sum_{i\in\mathcal{A}_{n'k}} t_i s_i/[\mu_{g(i)}(1+t_i)]$. Substituting (2) into (6),

$$X_n = w_n L_n + \sum_{n'} \mathcal{M}_{nn'} X_{n'}, \qquad \mathcal{M}_{nn'} = \sum_k \beta_k\Big[\sum_{g:\,k(g)=k} \theta_{gn} S_{g,n'} m(S_{g,n'}) + \mathbf{1}\{n'=n\}\,\tau_{nk}\Big] \geq 0.$$

Using $\sum_n \theta_{gn} = 1$, the column of $\mathcal{M}$ for n' sums to $\sum_k \beta_k[\sum_g S_{g,n'}m(S_{g,n'}) + \tau_{n'k}]$. Per dollar of spending in a market, revenue splits into profit, tariff revenue and labor cost, $\sum_g S_g m(S_g) + \tau + \sum_i s_i/[\mu_{g(i)}(1+t_i)] = 1$, and the last term is positive. So every column of $\mathcal{M}$ sums to less than one, $\rho(\mathcal{M}) < 1$, and $X = (I-\mathcal{M})^{-1}wL \geq wL$ is the unique solution. □

B.2 Proof of Proposition 4. Normalize wages to the unit simplex and let $z_n(w) = W_n(w)/w_n - L_n$ be excess demand for labor. By Propositions 2–3, z is continuous on the interior of the simplex and homogeneous of degree zero. It satisfies Walras' law $\sum_n w_n z_n(w) = 0$. Adding (6) across countries, world income is $\sum_n w_n L_n$ plus world profits plus world tariff revenue. By (2), world spending equals world income, and by the revenue split in B.1 it also equals $\sum_n W_n$ plus world profits plus world tariff revenue. Hence $\sum_n W_n = \sum_n w_n L_n$. It is bounded below by $-L$. It remains

to check the boundary condition of Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1): if $w^m \to \bar{w}$, where $Z = \{n : \bar{w}_n = 0\}$ is non-empty, then $\max_n z_n(w^m) \to \infty$.

A plant's delivered cost is proportional to the wage where it is located. So the costs of plants located in Z go to zero, while all other costs stay in a compact subset of $(0, \infty)$. By hypothesis, some plant located in Z serves a market (n', k) with $n' \notin Z$. By B.1, $X_{n'} \geq w_{n'} L_{n'}$, so $E_{n'k} = \beta_k X_{n'}$ stays bounded away from zero. For m large, the lowest-cost plant i in that market is located in some $\ell \in Z$. Let g be its group and J the number of plants in the market. Its cost is $V_i = E_{n'k} s_i / [\mu_g (1+t)]$, all of it paid to labor in ℓ, so it suffices to show that $V_i / w_\ell \to \infty$.

If $S_g \leq \frac{1}{2}$, then $\mu_g \leq 2\sigma/(\sigma - 1)$. Every price is at least c_i, since markups exceed one, so $P^{1-\sigma} \leq J c_i^{1-\sigma}$ and $s_i = (\mu_g c_i / P)^{1-\sigma} \geq \mu_g^{1-\sigma} / J$. Then V_i is bounded away from zero while $w_\ell \to 0$.

If $S_g > \frac{1}{2}$, every rival h has $S_h \leq 1 - S_g < \frac{1}{2}$, so $(1 - S_h)^{1-\sigma} \leq 2^{\sigma-1}$ and $1/S_g < 2$. By (10),

$$\frac{K_h}{K_g} = \frac{\psi(S_h)}{\psi(S_g)} = \frac{S_h (1 - S_h)^{1-\sigma}}{S_g (1 - S_g)^{1-\sigma}} \leq 2^\sigma (1 - S_g)^\sigma,$$

so $1 - S_g \geq \frac{1}{2} (K_h / K_g)^{1/\sigma}$. Rival strength K_h is bounded below, because all costs are bounded above. Plant i is the cheapest in its group, so $K_g \leq J c_i^{1-\sigma}$ and $s_i \geq S_g / J > 1/(2J)$. With $1/\mu_g = (\sigma - 1)(1 - S_g)/\sigma$, this gives $V_i \geq c\, c_i^{(\sigma-1)/\sigma}$ for some $c > 0$, and since c_i is proportional to w_ℓ, V_i / w_ℓ grows like $w_\ell^{-1/\sigma}$.

In either case $z_\ell \to \infty$, and the result follows from Mas-Colell, Whinston, and Green (1995, Proposition 17.C.1). □

B.3 A decomposition of labor payments. Raise wages, holding tariffs and trade costs fixed, and let hats denote log changes. A plant's delivered cost changes by $\hat{c}_i = \hat{w}_{\ell(i)}$. Its cost in a market is $V_i = r_i / [\mu_g (1 + t)]$, so $\hat{V}_i = \hat{E} + \hat{s}_i - \hat{\mu}_g$ with $\hat{s}_i = (1 - \sigma)(\hat{c}_i + \hat{\mu}_g - \hat{P})$ and $\hat{\mu}_g = \varepsilon_\mu \hat{S}_g$, where $\varepsilon_\mu = S_g / (1 - S_g)$ is the elasticity of the markup. By Proposition 5, $\hat{S}_g = -(\sigma - 1)\Lambda_g (\hat{c}_g - \hat{P})$, and using $1 = \Lambda_g [1 + (\sigma - 1)\varepsilon_\mu]$,

$$\hat{V}_i = \hat{E} - (\sigma - 1)(\hat{c}_i - \hat{c}_g) - \chi_g (\hat{c}_g - \hat{P}), \qquad \chi_g \equiv (1 - \varepsilon_\mu)(\sigma - 1)\Lambda_g = \frac{(\sigma - 1)(1 - 2S_g)}{1 + (\sigma - 2) S_g}. \quad \text{(B.3)}$$

The first term is the growth of the market, the second the reallocation of sales within the group, the third the reallocation between the group and its rivals. The coefficient χ_g lies in $[0, \sigma - 1]$ exactly when $S_g \leq \frac{1}{2}$.

B.4 Proof of Theorem 1. *The bound.* Raise w_j alone, $j \neq n$. Every plant paying country n is located in n, so $\hat{c}_i = 0$ for those plants. Every other plant has $\hat{c}_i \in \{0, 1\}$, so $\hat{c}_g \in [0, 1]$ and, by Proposition 5, $\hat{P} = \sum_g \varpi_g \hat{c}_g \in [0, 1]$. With $\hat{c}_i = 0$ the last two terms of (B.3) equal $(\sigma - 1)\hat{c}_g - \chi_g (\hat{c}_g - \hat{P})$. If $\hat{c}_g \geq \hat{P}$, condition (i) gives $\chi_g \leq \sigma - 1$ and the expression is at least $(\sigma - 1)\hat{P} \geq 0$; otherwise both parts are non-negative, since $\chi_g \geq 0$. So $\hat{V}_i \geq \hat{E}$ for every plant paying country n, and weighting by $b^n_{i,n'}$, which sum to one, gives $M_{nj} \geq \bar{E}^{(j)}_n$. *Uniqueness.* The argument is classical (Arrow, Block, and Hurwicz, 1959); see also Mas-Colell, Whinston, and Green (1995, Section 17.F). Since M is homogeneous of degree zero in wages, conditions (i) and (ii) also hold at every rescaling of a point of D, and the set of these rescalings is still convex in logarithms. Let $w, w^* \in D$ be non-proportional equilibria. Scale w^* so that $w^* \leq w$ with equality at some n_0, and move from w^* to w along the segment in logarithms. Every wage other than w_{n_0} weakly rises and at least one strictly rises, so W_{n_0} strictly rises while w_{n_0} does not. Then $z_{n_0}(w) > z_{n_0}(w^*) = 0$, a contradiction. □

C Comparative Statics

C.1 Proof of Proposition 5. By (9), $d\ln K_g = (1-\sigma)\hat{c}_g$, and $d\ln A = (1-\sigma)\,d\ln P$. Differentiating $\psi(S_g) = K_g/A$, with $d\ln\psi/d\ln S = 1+(\sigma-1)\varepsilon_\mu = 1/\Lambda_g$, gives $d\ln S_g = -(\sigma-1)\Lambda_g(\hat{c}_g - d\ln P)$. Shares sum to one, so $\sum_g S_g\, d\ln S_g = 0$, which gives $d\ln P = \sum_g \varpi_g \hat{c}_g$. If only j's costs change, $d\ln P = \varpi_j\hat{c}_j$, so $d\ln S_j = -(\sigma-1)\Lambda_j(1-\varpi_j)\hat{c}_j$ and, for $h \neq j$, $d\ln S_h = (\sigma-1)\Lambda_h\varpi_j\hat{c}_j$. Prices are $p_i = \mu(S_g)c_i$, so $d\ln p_i = d\ln c_i + \varepsilon_\mu(S_g)\,d\ln S_g$. Averaging over group j's plants with weights s_i/S_j gives $\rho_j\hat{c}_j$, and every rival's price rises. Since $(\sigma-1)\varepsilon_\mu\Lambda_j < 1$ and $1-\varpi_j \in [0,1)$, $\rho_j \in (0,1]$. □

C.2 Proof of Proposition 6. By (12), $\pi^H/E = \sigma^{-1}\sum_g \theta_g S_g(1+\kappa S_g) = \sigma^{-1}[\bar{\theta} + \kappa\sum_g \theta_g S_g^2]$, and $\sum_g \theta_g S_g^2 = \bar{\theta}\cdot\mathrm{HHI} + \mathrm{Cov}_S(\theta, S)$ by definition. Total profit is the case $\theta_g \equiv 1$, $\sum_g \pi_g/E = (1+\kappa\,\mathrm{HHI})/\sigma$, using $\sum_g S_g = 1$. Subtracting $\bar{\theta}\sum_g \pi_g$ from π^H leaves $(E\kappa/\sigma)\mathrm{Cov}_S(\theta, S)$. □

C.3 Proof of Proposition 7. By (12), $d\pi_g = (E/\sigma)(1+2\kappa S_g)\,dS_g$. Let $Z = \sum_h S_h\Lambda_h$, so $\varpi_j = S_j\Lambda_j/Z$. From C.1, $dS_j = -(\sigma-1)S_j\Lambda_j(1-\varpi_j)\hat{c}_j = -(\sigma-1)\varpi_j\hat{c}_j\sum_{h\neq j} S_h\Lambda_h$ and $dS_h = (\sigma-1)\varpi_j S_h\Lambda_h\hat{c}_j$ for $h \neq j$. Substituting into $d\pi^H = \sum_g \theta_g d\pi_g$ and collecting terms gives (18). □

D Entry

D.1 Two elasticities. Let $G(S) \equiv \pi'(S)/\psi'(S)$, where $\pi(S) = E(S+\kappa S^2)/\sigma$, and define $\varepsilon_G = SG'/G$ and $\varepsilon_\psi = S\psi'/\psi$. Differentiating $\psi(S) = K/A$ gives $\partial S/\partial K = 1/(A\psi')$ and $\partial S/\partial A = -\psi/(A\psi')$. Hence $\Omega_K = G(S)/A > 0$, $\Omega_{KK} = G'(S)/(A^2\psi'(S))$, and

$$\Omega_{KA} = -\frac{1}{A^2}\Big[G'(S)\frac{\psi(S)}{\psi'(S)} + G(S)\Big] = -\frac{G(S)}{A^2\varepsilon_\psi(S)}\big[\varepsilon_G(S) + \varepsilon_\psi(S)\big].$$

So Ω is strictly concave in K if and only if $\varepsilon_G < 0$, and $\Omega_{KA} < 0$ if and only if $\varepsilon_G + \varepsilon_\psi > 0$.

D.2 Proof of Theorem 2(a)–(b). Set $E = 1$, $a \equiv 2(\sigma-1)$ and $b \equiv \sigma - 2 \geq 0$. Then $\pi'(S) = (1+aS)/\sigma$ and $\psi'(S) = C(1-S)^{-\sigma}(1+bS)$ with $C = [\sigma/(\sigma-1)]^{\sigma-1}$, so

$$\varepsilon_G(S) = S\,f(S), \qquad f(S) = \frac{a}{1+aS} - \frac{b}{1+bS} - \frac{\sigma}{1-S}.$$

At $S = 0$, $f(0) = a - b - \sigma = 0$. Since $a > b \geq 0$, $a/(1+aS) > b/(1+bS) \geq 0$, so $f'(S) = -a^2/(1+aS)^2 + b^2/(1+bS)^2 - \sigma/(1-S)^2 < 0$. Hence $f < 0$ and $\varepsilon_G < 0$ on $(0,1)$, which is (a). Next, $\varepsilon_\psi = (1+bS)/(1-S)$, and using $1 + bS - \sigma S = 1 - 2S$,

$$\varepsilon_G + \varepsilon_\psi = \frac{aS}{1+aS} - \frac{bS}{1+bS} + \frac{1-2S}{1-S}.$$

The map $x \mapsto x/(1+x)$ is increasing and $a > b$, so the first two terms sum to a positive number; the third is non-negative when $S \leq \frac{1}{2}$. This is (b). At $S = \frac{1}{2}$ the expression equals $(\sigma-1)/\sigma - (\sigma-2)/\sigma = 1/\sigma$. □

D.3 Proof of Theorem 2(c). *Step 1: the ladder.* Since $\Omega_K > 0$, a set T is optimal only if no T' has $K(T') \geq K(T)$ and $\Phi(T') \leq \Phi(T)$ with one inequality strict. The remaining efficient sets can be ordered so that K and Φ rise together. *Step 2: single crossing.* Take $A < A'$ and two rungs with $K_1 < K_2$, and let $\Delta(x) = \int_{K_1}^{K_2}\Omega_K(K;x)\,dK - [\Phi_2 - \Phi_1]$ be the net gain from the larger rung at aggregate x. The share $\psi^{-1}(K/x)$ rises with K and falls with x, so on $[K_1, K_2]\times[A, A']$ it is largest

at (K_2, A). If $\psi^{-1}(K_2/A) \leq \frac{1}{2}$, part (b) gives $\Omega_{KA} < 0$ throughout, and Δ is strictly decreasing on $[A, A']$. *Step 3: the aggregate.* Given strengths, the aggregate $A(\{K_g\})$ solving (11) is strictly increasing in each K_g: raising K_j at a given A raises the left side, which is strictly decreasing in A. *Step 4.* Suppose $(A, \{K_g\})$ and $(A', \{K'_g\})$ are equilibria with $A < A'$, and suppose $K'_g > K_g$ for some g. Apply Step 2 with $K_1 = K_g$ and $K_2 = K'_g$: by the hypothesis, group g would hold at most half of the market with the plants it chooses at A' while facing A, so $\psi^{-1}(K'_g/A) \leq \frac{1}{2}$ and Δ is strictly decreasing on $[A, A']$. Optimality at A' gives $\Delta(A') \geq 0$, hence $\Delta(A) > 0$, so at A the group strictly prefers K'_g to K_g, contradicting optimality at A. Hence $K'_g \leq K_g$ for every g (the single-crossing argument of Milgrom and Shannon, 1994), and by Step 3, $A' = A(\{K'_g\}) \leq A(\{K_g\}) = A$, a contradiction. Equilibria with the same aggregate have the same best responses, and with no ties the same plant sets. □

D.4 The notion of entry equilibrium. Definition 2 evaluates every alternative set at the prevailing aggregate A. In a two-stage game in which the market is re-solved after a deviation, a group that adds a plant anticipates that A rises, which lowers the value of the deviation; the two notions coincide for groups with negligible shares and can differ for large ones. Theorem 2 applies to the notion of Definition 2. Existence in pure strategies is not guaranteed under either notion: a group's choice is discrete, so its optimal strength jumps with A, and (11) may have no solution with each group on a rung.